MODDIV-NOISE

Non-Cyclic Hankel Rounding Encryption, Exact Joint Simulation, and Conditional IND-CPA Security

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Abstract

We define NOISE-CPA-LIFT, a public-key encryption instantiation built around the MODDIV-NOISE public observation. The construction uses a non-cyclic public coefficient sequence with independent entries, secrets uniform in a positive box, and modular reduction followed by coefficientwise integer division. Its explicit noise has the form $2e + \tau$, where e follows a fixed distribution and τ follows the centered binomial distribution with parameter one. Encryption uses an internal randomized lift of the public quotient and two Hankel products sharing one ephemeral secret.

We give a deterministic sufficient correctness condition for bounded noise and a failure bound for noise with nonzero tails. The proof retains the actual division remainders and uses circular distance. We then prove a classical standard-model IND-CPA reduction to an explicitly defined decisional binary-secret Hankel-LWE problem. Each of two exact simulations uses a single source coefficient sequence; the second preserves the joint distribution of the ciphertext components by selecting two disjoint public segments with a shared secret. The manuscript provides exact compact-encoding sizes, a Python reference implementation, and reproducible checks. Hardness of the Hankel-LWE assumption is not derived here from standard LWE or MP-LWE. No concrete security level or deployment parameters are proposed. The unwrapped encryption scheme is malleable and is not IND-CCA2 secure.

Keywords: ModDiv; Toeplitz; Hankel; public-key encryption; structured LWE; rounding; randomized lifting; IND-CPA.

Scope. This article presents a construction and a conditional proof. It is neither a security certification nor a retroactive proof for historical MODDIV. The positive theorem is conditional on the explicitly defined binary-secret Hankel-LWE source; no reduction from standard LWE, MP-LWE, or worst-case lattice problems is claimed here. The unwrapped scheme is not IND-CCA2 secure, and no concrete 128-bit security level is asserted.

Contents

1 Introduction and contributions **3**

2 Related work and positioning **4**

3 Preliminaries and structured assumption **4**

4 The NOISE-CPA-LIFT construction **7**

5 Correctness with the actual remainders **8**

6 Distribution-simulation lemmas **10**

7 Conditional IND-CPA security **13**

8 Sizes, costs, and reproducible verification **14**

9 Cryptographic limitations and open obligations **17**

10 Conclusion **18**

Assistance disclosure and availability **18**

A Complete pseudocode for the two reductions **20**

B Decoding, noise, and encoding details **21**

C Reproducibility package guide **21**

Main result. Under the decisional assumption

$$\mathsf{HB}\big(M, d, 2(d+\ell-1), \chi\big),$$

the construction specified here admits a classical standard-model IND-CPA reduction with two exact transitions. Correctness additionally requires a sufficient decoding margin. A reduction to this structured assumption does not establish the hardness of the assumption itself.

1 Introduction and contributions

1.1 From rounded public observations to encryption

A public observation of the form "structured product, modular reduction, then integer division" defines an observation problem. It does not, by itself, define encryption: one must specify how a message is embedded, how the secret-key holder recovers it, and which experiment an adversary must distinguish. This manuscript makes that separation for a noisy variant of ModDiv.

Bouftass's multidimensional proposal [1] uses truncated polynomial products and modulo–division operations. We retain its non-cyclic representation framework, but do not inherit a historical security claim. The *MODDIV-NOISE* public observation studied here is fully specified by uniform distributions and explicit noise. The name NOISE-CPA-LIFT identifies the added encryption layer: an ephemeral secret, a randomized lift of the public observation, independent final noise, and two-center bit encoding.

The technical point is a joint simulation. The two ciphertext components share the same ephemeral secret. Replacing them separately by objects with the correct marginals would not establish their joint distribution. Our second simulator takes two disjoint segments from one long Hankel coefficient sequence, uses the same source secret and the same mask, and discards exactly the windows that cross the boundary.

1.2 Results established

Three public, efficient algorithms specify the construction. For a source error supported on $[-b, b]$, correctness is guaranteed by

$$[2d(Q-1)+1](Q+2b) < P/4. \tag{1.1}$$

This is a sufficient bound, not a claim of optimality. It does not assume independent rounding remainders and remains valid near wraparound modulo P.

For every CPA adversary $\mathcal{A}$, we construct two distinguishers $\mathcal{B}_0, \mathcal{B}_1$ for the same binary-secret HB-LWE problem such that

$$\mathrm{Adv}_{\mathrm{CPA}}(\mathcal{A}) \le \mathrm{Adv}_{\mathsf{HB}}(\mathcal{B}_0) + \mathrm{Adv}_{\mathsf{HB}}(\mathcal{B}_1). \tag{1.2}$$

The simulations are exact under the stated distributions. They use no random oracle, exponential preprocessing, or access to the source-instance secret. The bound involves two conceptually separate challenges in the hybrid argument, not a simulator that reuses a challenge as though it could resubmit it under another distribution.

We also give a lossless encoding, a corollary on the secrecy of a uniformly chosen key that is then encrypted, and an explicit CCA2 attack against the unwrapped scheme. The programs verify identities, selected complete finite games, encodings, and correctness instances. They check the specification; they do not prove hardness.

1.3 Scope of the security claim

The starting assumption is *non-cyclic binary-secret Hankel-LWE*, with a finite number of windows determined by the encryption dimensions. It is not another name for LWE with a general matrix, nor a consequence of MP-LWE theorems invoked without proof. The word "conditional" in the title is part of the scientific result.

Three obligations must be distinguished: decryption correctness, reduction of the CPA game to HB-LWE, and justification of HB-LWE hardness for a useful parameter family. The first two are addressed here. The third remains to be established. A functional parameter family may satisfy correctness without any confidentiality justification. No key size reported in this text is associated with a demonstrated 128-bit security level.

1.4 Contribution and novelty

Middle products, Toeplitz and Hankel matrices, randomized quotient lifting, and game-based proofs have precedents. We do not claim to have invented these tools. The precise object submitted for scrutiny is their instantiation with the distributions of NOISE-CPA-LIFT, the exact binary-to-positive mask, and the joint simulation of the ideal key and ciphertext. The overall novelty of this combination requires an independent comparison with the literature and external review.

2 Related work and positioning

2.1 LWE and binary secrets

Regev [2] relates worst-case lattice problems to a form of LWE and constructs encryption from that assumption under the conditions of his result. Micciancio [3] gives a reduction to binary-secret LWE with a polynomial number of samples, controlling dimensions and noise growth. These results motivate noisy problems with binary secrets; they are not applied automatically to this manuscript's structured coefficient sequence.

In general LWE, the public matrix entries are independent. In our assumption, several rows are overlapping windows of a single sequence. Preserving the bilinear equation therefore does not suffice to preserve the statistical experiment. Likewise, a binary secret support does not determine the error distribution: a CBD1 instantiation of the code does not, for that reason alone, inherit a theorem for other parameters.

2.2 Middle products and randomized lifting

Roşca, Sakzad, Stehlé, and Steinfeld [4] introduce MP-LWE and CPA encryption, with a reduction from Polynomial-LWE under their conditions. The identity between a polynomial-product slice and a structured matrix is a direct precedent for the language used here.

Bai et al. [5] study MP-CLWR and rounding-based encryption in the random-oracle model. Their probabilistic function INV selects a preimage of a quotient. Our $B = QU + \rho$ is an elementary instance of this idea for equal-size cells and floor rounding. The present proof does not use their computational assumption, random oracle, or parameters; it establishes from scratch the distributions and simulation required here.

2.3 Algebraic structures and tensors

Peikert and Pepin [6] unify several algebraic LWE variants using bilinear maps and tensors. Their presentation identifies, in particular, the Hankel slices of the middle-product tensor. It emphasizes that a reduction must transport spaces, bases, secrets, and errors, not merely the coefficient pattern. We do not claim that one of their theorems directly covers $\mathsf{HB}(M, d, N, \chi)$ with our binary secrets and windows.

2.4 Comparison of proof commitments

3 Preliminaries and structured assumption

3.1 Arithmetic and probabilistic notation

For $t \geq 2$, we identify $\mathbb{Z}_t$ with $\{0, \ldots, t-1\}$ for encoding. The notation $[z]_t$ denotes the canonical residue of an integer z. Reduction and division act coordinatewise on vectors. Integer intervals

Table 1: Positioning of the proofs; parameter claims from the references are not imported.

Setting	Relevant object	Distinction in this work
General LWE [2, 3]	Noisy equations, hardness reductions, binary secrets	The public object is a general matrix, not a single Hankel sequence.
MP-LWE [4]	Polynomial middle product, CPA encryption	The roles, dimensions, and distributions of the factors require an explicit connection.
MP-CLWR [5]	Rounding, randomized lifting, computational proof	The cited theorem does not establish our decisional HB assumption.
Algebraic LWE [6]	Bilinear representation and transports between structures	A tensor identity does not establish the required binary-secret distribution.
Present scheme	Floor rounding, positive secrets, noise $2e + \tau$	Two exact simulations from an explicit HB assumption; external hardness is not established.

$[a, b)$ include a and exclude b.

The notation $x \leftarrow\$ \mathcal{D}$ means sampling from distribution $\mathcal{D}$. The uniform distribution on a finite set E is $\mathsf{U}(E)$. A tensor power denotes independent sampling. Statistical distance is

$$\mathrm{TV}(P_1, P_0) = \frac{1}{2} \sum_z |P_1(z) - P_0(z)|.$$

A common randomized post-processing cannot increase it: each output acceptance probability is an average of conditional acceptance probabilities between zero and one. We use this property only to account for sampler approximations, not as a substitute for a hardness reduction.

Circular distance modulo P is

$$\mathrm{dist}_P(x, y) = \min_{k \in \mathbb{Z}} |x - y - kP|.$$

All decoding conditions use this distance. An ordinary-distance bound between canonical representatives would generally fail when wrapping from $P - 1$ to zero.

3.2 Hankel products, Toeplitz matrices, and polynomials

For a coefficient sequence a of length $r + d - 1$, define

$$\mathsf{H}_r(a)_{i,j} = a_{i+j}, \qquad (\mathsf{H}_r(a)x)_i = \sum_{j=0}^{d-1} a_{i+j} x_j, \tag{3.1}$$

where $0 \le i < r$ and $0 \le j < d$. There is no cyclic wraparound in the indices. The number of columns is always the length of the multiplier vector.

If J_d reverses the coordinates, $\mathsf{H}_r(a)J_d$ is Toeplitz: its (i, j) entry is $a_{i+d-1-j}$ and depends only on $i - j$. This explains the Toeplitz–Hankel terminology in the title without introducing two different public distributions.

For $a(z) = \sum_k a_k z^k$ and $x^\star(z) = \sum_{j=0}^{d-1} x_j z^{d-1-j}$,

$$[z^{d-1+i}]\, a(z)x^\star(z) = (\mathsf{H}_r(a)x)_i. \tag{3.2}$$

This representation is a slice of an ordinary convolution. It does not involve a quotient by $z^d - 1$ or $z^d + 1$.

3.3 Parameters and distributions

Fix

$$P = 2^p = 2M, \quad Q = 2^q, \quad 3 \le q < p, \quad H = P/Q, \quad d \ge 2, \quad \ell \ge 2. \tag{3.3}$$

The ordinary letter H denotes the quotient-space cardinality; H_r denotes the Hankel matrix. In particular, H is even. Set

$$n = d + \ell - 1, \qquad m = 2d + \ell - 2 = n + d - 1, \qquad N = 2n. \tag{3.4}$$

Table 2: Dimensions used in the scheme and the proof.

Symbol	Value or domain	Role
d	integer ≥ 2	Length of the private and ephemeral secrets.
ℓ	integer ≥ 2	Binary message length.
n	$d + \ell - 1$	Length of U, V, and the lift B.
m	$n + d - 1$	Length of the public sequence A.
N	$2n$	Number of HB windows required by the reduction.
$N + d - 1$	$m + n$	Length of the long source sequence.
$\mathcal{I}_Q$	$[Q/2, Q) \cap \mathbb{Z}$	Alphabet of positive secrets.
F_Q	$\mathbb{Z}_P \to \mathbb{Z}_H$	$F_Q(z) = \lfloor [z]_P / Q \rfloor$.

Let χ be an integer distribution fixed publicly before key generation. Its sampler and the bit lengths of the generated integers are part of the specification of the family. Every explicit error in the scheme has distribution

$$\chi' = \mathcal{L}(2e + \tau), \qquad e \leftarrow\$ \, \chi, \quad \tau \leftarrow\$ \, \mathsf{CBD}_1, \tag{3.5}$$

with independence, where

$$\Pr[\tau = -1] = \Pr[\tau = 1] = \frac{1}{4}, \qquad \Pr[\tau = 0] = \frac{1}{2}.$$

Each error occurrence uses fresh randomness. The proofs assume neither symmetry nor Gaussianity of χ. The correctness bounds add a support or tail condition. The Python reference implements only $\chi = \mathsf{CBD}_1$; its samplers draw two fair bits and take their difference.

3.4 The binary-secret HB-LWE experiment

Definition 3.1 (The two HB worlds)**.** For (M, d, N, χ), independently draw

$$a \leftarrow\$ \, \mathsf{U}(\mathbb{Z}_M^{N+d-1}), \quad S \leftarrow\$ \, \mathsf{U}(\{0,1\}^d), \quad e \leftarrow\$ \, \chi^{\otimes N}.$$

The real world HB_1 returns (a, b) with

$$b = [\mathsf{H}_N(a) S + e]_M.$$

The null world HB_0 returns (a, b) with $b \leftarrow\$ \, \mathsf{U}(\mathbb{Z}_M^N)$ independent of a. For a distinguisher $\mathcal{B}$, define

$$\mathrm{Adv}_{\mathsf{HB}}(\mathcal{B}) = \left| \Pr[\mathcal{B}(\mathsf{HB}_1) = 1] - \Pr[\mathcal{B}(\mathsf{HB}_0) = 1] \right|.$$

The secret is shared by all windows. The distinguisher receives one sequence and its N observations; no additional observations, adaptive public choices, or repetitions under the same secret are supplied implicitly. Errors are reduced modulo M in the observations, but retain their stated integer distribution for the correctness analysis after lifting.

Definition 3.2 (The conditional assumption used in this work)**.** For a publicly specified family par(λ) whose dimensions and descriptions are polynomial in λ, the above advantage is negligible for every classical probabilistic polynomial-time distinguisher, with $N = 2(d + \ell - 1)$ and the chosen distribution χ.

This assumption contains neither an encrypted message nor an encapsulated key. It is therefore distinct from the CPA statement it supports. Nevertheless, this work does not establish it through a reduction from general LWE. Its window count is a commitment: hardness for n windows does not justify assuming hardness for the $2n$ windows used here at no cost.

4 The NOISE-CPA-LIFT construction

The parameters are public and fixed. The message space is $\{0,1\}^\ell$. The algorithms below operate on integer vectors; their canonical encoding is defined in section 8.

4.1 Key generation

Algorithm 1. $\mathsf{KeyGen}(\mathrm{par})$

Input: Parameters satisfying (3.3) and distribution χ.
$A \leftarrow\$\ \mathsf{U}([M,P)^m)$; $X \leftarrow\$\ \mathsf{U}(\mathcal{I}_Q^d)$;
$E'_0 \leftarrow\$\ (\chi')^{\otimes n}$, independently;
$U \leftarrow F_Q(\mathsf{H}_n(A)X + E'_0)$;
return $pk = (A, U)$ *and* $sk = X$.

The public key contains the complete sequence, not its expanded matrix and not an expansion seed. The upper-half domain is exact: the most significant bit of each A_i is fixed to one. The secret key does not include the key-generation errors, which are unnecessary for decryption.

4.2 Encryption

Algorithm 2. $\mathsf{Enc}(pk, \mu)$

Input: A valid $pk = (A, U)$ and $\mu \in \{0,1\}^\ell$.
$Y \leftarrow\$\ \mathsf{U}(\mathcal{I}_Q^d)$;
$E'_1 \leftarrow\$\ (\chi')^{\otimes n}$; $E'_2 \leftarrow\$\ (\chi')^{\otimes \ell}$;
$\rho \leftarrow\$\ \mathsf{U}(\{0,\ldots,Q-1\}^n)$, with all draws fresh and independent;
$B \leftarrow QU + \rho$;
$V \leftarrow F_Q(\mathsf{H}_n(A)Y + E'_1)$;
$W \leftarrow F_Q(\mathsf{H}_\ell(B)Y + E'_2 + (P/2)\mu)$;
return $ct = (V, W)$.

The lift B is not published. Its length is $n = \ell + d - 1$, allowing the second product. Both products use exactly the same Y. Drawing two independent ephemeral secrets would change the construction and destroy the correctness identity. By contrast, E'_1, E'_2, and ρ are fresh and independent.

The value B belongs to $\mathbb{Z}_P^n$, but is not assumed uniform in the real system. Its uniformity is used only in a proof game after replacing U. The choice $B = QU$ without a random remainder does not have this property.

4.3 Decryption and formats

Algorithm 3. $\mathsf{Dec}(sk, ct)$

Input: $sk = X$ and $ct = (V, W)$.
Check $X \in \mathcal{I}_Q^d$, $V \in \mathbb{Z}_H^n$, and $W \in \mathbb{Z}_H^\ell$; otherwise reject;
$Z \leftarrow [QW - \mathsf{H}_\ell(QV)X]_P$;
for $i = 0, \ldots, \ell - 1$ **do**
  $\widehat{\mu}_i \leftarrow 1$ if $\mathrm{dist}_P(Z_i, P/2) < \mathrm{dist}_P(Z_i, 0)$; otherwise $\widehat{\mu}_i \leftarrow 0$;
return $\widehat{\mu}$.

A distance tie is decoded as zero; honest ciphertexts cannot produce a tie when the strict correctness margin holds. Invalid formats are rejected, but this check provides no CCA security. Ideal keys used in the CPA proof need not have an associated secret key: no decryption oracle is invoked in those games.

4.4 Explicit changes relative to the original public observation

Public-key generation preserves the positive box, upper-half public sequence, non-cyclic structure, floor rounding, and noise $2e + \tau$. Encryption adds four mechanisms: a positive ephemeral secret, a fresh lift $QU + \rho$, independent final noise with the same stated distribution, and message encoding at the centers $0, P/2$. There is no additional public reconciliation aid or ideal hash. Two-center decoding is the reconciliation mechanism of this instantiation.

5 Correctness with the actual remainders

5.1 The double-product identity

Lemma 5.1 (Commutation of middle products)**.** *For $A \in \mathbb{Z}^{2d+\ell-2}$ and $X, Y \in \mathbb{Z}^d$, setting $n = d + \ell - 1$, the following identity holds over the integers:*

$$\mathsf{H}_\ell(\mathsf{H}_n(A)X)Y = \mathsf{H}_\ell(\mathsf{H}_n(A)Y)X. \tag{5.1}$$

The same identity holds after reduction modulo P.

Proof. Coordinate i of the left-hand side equals

$$\sum_{k=0}^{d-1} \sum_{j=0}^{d-1} A_{i+k+j} X_j Y_k.$$

Exchanging the two summation indices gives the right-hand side. For $0 \le i < \ell$, the largest index of A is $\ell - 1 + 2d - 2 = m - 1$; all coordinates used exist. No periodicity is assumed. □

5.2 Exact error expansion

Introduce the remainders actually generated by the operations:

$$\begin{aligned} r_0 &= [\mathsf{H}_n(A)X + E_0']_Q, \\ r_1 &= [\mathsf{H}_n(A)Y + E_1']_Q, \\ r_2 &= [\mathsf{H}_\ell(B)Y + E_2']_Q. \end{aligned} \tag{5.2}$$

Since $Q \mid P$ and $Q \mid P/2$, these are the remainders of the modular residues used by the algorithms. The last one does not depend on μ. They lie in the appropriate-dimensional boxes $[0, Q)$, but are generally neither independent of the secrets nor uniform.

Lemma 5.2 (Decryption residue)**.** *For an honest encryption, the decryption residue satisfies*

$$Z \equiv (P/2)\mu + \Delta \pmod{P}, \tag{5.3}$$

with the integer lift

$$\Delta = \mathsf{H}_\ell(E_0' - r_0 + \rho)Y - \mathsf{H}_\ell(E_1' - r_1)X + E_2' - r_2. \tag{5.4}$$

Proof. The definition of floor rounding gives the congruences

$$\begin{aligned} QU &\equiv \mathsf{H}_n(A)X + E_0' - r_0, \\ QV &\equiv \mathsf{H}_n(A)Y + E_1' - r_1, \\ QW &\equiv \mathsf{H}_\ell(B)Y + E_2' + (P/2)\mu - r_2. \end{aligned}$$

Replace B by $QU + \rho$, expand the products, and subtract $\mathsf{H}_\ell(QV)X$. The two principal terms cancel by lemma 5.1. The remaining terms are exactly those in (5.4). No probabilistic argument is used. □

5.3 Bounded noise

Theorem 5.3 (Perfect correctness under a sufficient margin)**.** *Suppose* $\mathrm{Supp}(\chi) \subseteq [-b, b] \cap \mathbb{Z}$*, with* $b \geq 0$*. Set*

$$T_b = [2d(Q-1)+1](Q+2b). \tag{5.5}$$

If $T_b < P/4$*, every honest ciphertext decrypts correctly, for every public sequence, admissible secret, and message.*

Proof. Each E_i' is bounded in absolute value by $2b+1$. Since the coordinates of r_0, r_1, r_2, ρ lie between zero and $Q-1$,

$$\|E_0' - r_0 + \rho\|_\infty \leq Q + 2b, \quad \|E_1' - r_1\|_\infty \leq Q + 2b, \quad \|E_2' - r_2\|_\infty \leq Q + 2b.$$

The coordinates of X, Y are at most $Q-1$. Each coordinate of each of the two products in (5.4) contains d terms. The triangle inequality gives $\|\Delta\|_\infty \leq T_b$.

The centers zero and $P/2$ are separated by distance $P/2$ on the circle. A residue at distance strictly less than $P/4$ from the correct center is strictly closer to it than to the other center. By (5.3), all bits are therefore recovered. The conclusion remains valid when an integer lift crosses zero or P. □

5.4 Noise tails and averaging over keys

Set $\delta_b = \Pr_{e\sim\chi}[|e| > b]$. One key generation and one encryption use $2n + \ell = 2d + 3\ell - 2$ draws from χ. If $T_b < P/4$, a union bound over tail events gives

$$\mathbb{E}_{pk,sk}\left[\max_{\mu\in\{0,1\}^\ell} \Pr[\mathsf{Dec}(sk, \mathsf{Enc}(pk, \mu)) \neq \mu \mid pk, sk]\right] \leq \min\{1, (2d + 3\ell - 2)\delta_b\}. \tag{5.6}$$

To justify the maximum, couple encryptions of all messages using the same coins: the event containing any failure depends only on the errors, not on the message. It can therefore be bounded before averaging over keys. This does not assert the same failure probability for every arbitrarily fixed key. For several honest encryptions, another union bound may be applied to fresh errors and reused key errors, without declaring the latter independent anew.

Remark 5.4 (Correctness is not confidentiality). For $\chi = \mathsf{CBD}_1$, take $b = 1$. The profile $(P, Q, d, \ell) = (4096, 8, 4, 2)$ satisfies $T_1 = 570 < 1024 = P/4$. This certifies correctness, not hardness. The theorem imposes no rank assumption on the public sequence.

6 Distribution-simulation lemmas

6.1 Exact binary-secret masking

Define the public set

$$\mathcal{R}_Q = \{Q/2 + 4t + \epsilon : \ 0 \le t < Q/8, \ \epsilon \in \{0,1\}\}. \tag{6.1}$$

Its cardinality is $Q/4$. The condition $Q \ge 8$ ensures a positive integer number of blocks.

Lemma 6.1 (Binary-to-positive bijection)**.** *If $R \leftarrow\$ \mathsf{U}(\mathcal{R}_Q^d)$ and $S \leftarrow\$ \mathsf{U}(\{0,1\}^d)$ are independent, then $R + 2S$ is exactly uniform on $\mathcal{I}_Q^d$.*

Proof. In one coordinate, every integer in $\mathcal{I}_Q$ has a unique representation

$$Q/2 + 4t + \epsilon + 2s, \qquad 0 \le t < Q/8, \quad \epsilon, s \in \{0,1\}.$$

The pair (ϵ, s) ranges over the four residues of its block. The map $(R, s) \mapsto R + 2s$ is a bijection between sets of cardinality $Q/2$. Taking the product of these bijections gives the vector distribution. □

6.2 Lifting with mixed public high bits

Lemma 6.2 (Transport of the two worlds)**.** *Let $a \in \mathbb{Z}_M^{r+d-1}$ and $b \in \mathbb{Z}_M^r$ be an HB challenge. Choose a vector $\sigma \in \{0,1\}^{r+d-1}$ independently of the secret and errors, conditional on a; its distribution may depend on a. Set $A = a + M\sigma$. Independently draw $R \leftarrow\$ \mathsf{U}(\mathcal{R}_Q^d)$ and $\tau \leftarrow\$ \mathsf{CBD}_1^{\otimes r}$, and compute*

$$C = [2b + \mathsf{H}_r(A)R + \tau]_P. \tag{6.2}$$

In the real world,

$$C = [\mathsf{H}_r(A)(R + 2S) + 2e + \tau]_P.$$

In the null world, C is uniform on $\mathbb{Z}_P^r$, even conditional on (a, σ, R).

Proof. Since $A - a = M\sigma$ and $P = 2M$, we have $2\mathsf{H}_r(A)S \equiv 2\mathsf{H}_r(a)S \pmod P$. Doubling a residue modulo M has the same value modulo $2M$ as doubling its integer lift. This proves the real-world identity. Lemma 6.1 gives the uniform positive secret, and the errors e, τ remain independent of the variables defining it.

In the null world, one coordinate of $2b$ is uniform among the even residues. If z is even, $2b + \tau = z$ has probability $(1/2)/M = 1/P$. If z is odd, the two values $\tau = 1, -1$ contribute a total probability of $(1/4 + 1/4)/M = 1/P$. The coordinates are independent. Translation by $\mathsf{H}_r(A)R$ preserves the uniform distribution, even conditional on all the stated variables. □

Choosing $\sigma = \mathbf{1}$ gives $A \leftarrow\$ \mathsf{U}([M, P)^{r+d-1})$. Choosing independent fair bits for σ gives a public sequence uniform on all of $\mathbb{Z}_P^{r+d-1}$. This distinction is necessary for the second simulator. In both cases, the resulting noise is exactly $2e + \tau$, not a new error distribution chosen to make a comparison easier.

Corollary 6.3 (The NOISE public observation)**.** *Applying F_Q to (6.2), with $\sigma = \mathbf{1}$, transports a binary HB challenge exactly to the public observation in algorithm 1 in the real world, and to a uniform quotient independent of the public sequence in the null world.*

Proof. The real world follows from lemma 6.2. In the null world, every quotient has exactly Q preimages in $\mathbb{Z}_P$, because $Q \mid P$. Rounding therefore maps the uniform distribution to the uniform distribution on $\mathbb{Z}_H$, coordinatewise. □

6.3 Uniform lifting of a quotient

Lemma 6.4 (Quotient, remainder, and deferred lifting)**.** *If $B \leftarrow\$ \mathsf{U}(\mathbb{Z}_P^n)$ and $U = F_Q(B)$, then the joint distribution of (U, B) is obtained by independently drawing*

$$U \leftarrow\$ \mathsf{U}(\mathbb{Z}_H^n), \qquad \rho \leftarrow\$ \mathsf{U}(\{0, \ldots, Q-1\}^n), \qquad B = QU + \rho.$$

For the conditional statement used below, assume additionally that a public variable A is independent of B. If an adversary sees only (A, U) and uses coins independent of (A, B) to choose its messages and retained state, then preparing B before those choices is distributionally equivalent to drawing a fresh conditional lift $QU + \rho$ afterwards.

Proof. The scalar map $B \mapsto (\lfloor B/Q \rfloor, B \bmod Q)$ is a bijection, with inverse $(U, \rho) \mapsto QU + \rho$. Applying it coordinatewise proves independent uniform quotient and remainder. Since A is independent of B, conditioning on (A, U) leaves $\rho = [B]_Q$ uniform on $\mathbb{Z}_Q^n$. The adversary's messages and retained state are randomized functions of (A, U) and independent coins, so conditioning further on them does not change this conditional remainder law. This proves the deferred-sampling statement. □

The lemma is applied only in the ideal-public-key game. It does not say that a real quotient generated by KeyGen is uniform, or that the actual remainders of key generation are independent of X. Moreover, B is not independent of U: the relation $F_Q(B) = U$ is deterministic.

6.4 First simulator: idealizing the public observation

Simulator $\mathcal{B}_0$ receives (a, b) from $\mathsf{HB}(M, d, N, \chi)$ with $N = 2n$. It takes the prefix $a_{0:m}$ and observations $b_{0:n}$, where index intervals exclude their upper endpoint. It sets

$$\begin{aligned} A &= a_{0:m} + M\mathbf{1}, \\ U &= F_Q(2b_{0:n} + \mathsf{H}_n(A)R + \tau_0), \end{aligned} \tag{6.3}$$

with $R \leftarrow\$ \mathsf{U}(\mathcal{R}_Q^d)$ and $\tau_0 \leftarrow\$ \mathsf{CBD}_1^{\otimes n}$. It gives $pk = (A, U)$ to the adversary. After receiving the two messages, it samples a challenge bit, publicly encrypts the corresponding message, and returns the bit indicating whether the adversary's answer is correct.

Corollary 6.3 shows that the real world reproduces the honest key and the null world the ideal key with a uniform quotient. In both cases, subsequent encryption is exactly the public algorithm. The simulator needs neither the source secret nor a secret key for the ideal key. Unused suffixes of the challenge are discarded and never published.

6.5 Second simulator: two segments, one shared secret

This step transports the joint ciphertext distribution. The source-sequence length is

$$N + d - 1 = 3d + 2\ell - 3 = m + n. \tag{6.4}$$

The simulator partitions its coefficients and selects its observations as follows:

$$a^{(A)} = a_{0:m}, \qquad a^{(B)} = a_{m:m+n}, \tag{6.5}$$

$$b^{(A)} = b_{0:n}, \qquad b^{(B)} = b_{m:m+\ell}. \tag{6.6}$$

The last window of the first block ends at coefficient $(n-1)+(d-1) = m-1$. The first window of the second block starts at coefficient m. The second block contains $N - m = \ell$ observations, the last of which uses coefficient $N+d-2 = m+n-1$. The two public segments are independent because they use disjoint coordinates of a uniform sequence. The real observations are *not* asserted to be independent without conditioning: they share the same S.

Table 3: Partition of the long challenge; index ranges in this table are inclusive.

Block	Public coefficients	Observations	Use
First	$0, \dots, m-1$	$0, \dots, n-1$	Public A and component V.
Boundary	Windows crossing the split	$n, \dots, m-1$	$d-1$ observations discarded.
Second	$m, \dots, m+n-1$	$m, \dots, N-1$	Lift B and component W.

Preparing the key before the messages. Draw $\epsilon \leftarrow\$\ \mathsf{U}(\{0,1\}^n)$ and set

$$A = a^{(A)} + M\mathbf{1}, \qquad B = a^{(B)} + M\epsilon, \qquad U = F_Q(B). \tag{6.7}$$

The public sequence A has the upper-half distribution. The vector B is uniform on $\mathbb{Z}_P^n$ and independent of A. This explicit independence is the hypothesis needed for the conditional part of lemma 6.4. The simulator publishes (A, U) and retains B.

By lemma 6.4, and using $A \perp B$, conditional on this ideal key the retained B has exactly the distribution of a fresh lift $QU + \rho$ even after adaptive message selection. Preparing it before the messages are chosen therefore does not alter the game. Additional encryptions performed publicly by the adversary use their own fresh remainders and cannot access the B retained for the challenge.

Answering the challenge. After receiving μ_0, μ_1, draw $\beta \leftarrow\$\ \mathsf{U}(\{0,1\})$, a single mask $R \leftarrow\$\ \mathsf{U}(\mathcal{R}_Q^d)$, and independent vectors $\tau_1 \sim \mathsf{CBD}_1^{\otimes n}$, $\tau_2 \sim \mathsf{CBD}_1^{\otimes \ell}$. Compute

$$\begin{aligned} C_1 &= [2b^{(A)} + \mathsf{H}_n(A)R + \tau_1]_P, \\ C_2 &= [2b^{(B)} + \mathsf{H}_\ell(B)R + \tau_2]_P, \\ V &= F_Q(C_1), \qquad W = F_Q(C_2 + (P/2)\mu_\beta). \end{aligned} \tag{6.8}$$

The simulator sends (V, W) and returns the adversary's success bit.

Proposition 6.5 (Exact joint simulation)**.** *In the real HB world, the key and challenge produced by* (6.7)–(6.8) *have the distribution of the encryption algorithm applied to a key whose U is uniform and independent of A. In the null HB world, the ciphertext is uniform on $\mathbb{Z}_H^n \times \mathbb{Z}_H^\ell$, independently of the key, the challenge bit, and the adversary's prior state.*

Proof. In the real world, the selected observations satisfy

$$b^{(A)} = [\mathsf{H}_n(a^{(A)})S + e^{(A)}]_M, \qquad b^{(B)} = [\mathsf{H}_\ell(a^{(B)})S + e^{(B)}]_M.$$

Their errors are independent because their indices are distinct; their secret is the same. Applying lemma 6.2 with the same R and the respective high bits gives

$$C_1 = [\mathsf{H}_n(A)Y + 2e^{(A)} + \tau_1]_P, \quad C_2 = [\mathsf{H}_\ell(B)Y + 2e^{(B)} + \tau_2]_P,$$

where $Y = R + 2S$ is uniform in the positive box and independent of A, B. These equalities hold simultaneously. Together with lemma 6.4, they show that all draws have the distribution of Enc in the ideal-key game, including the link through Y between the two products.

In the null world, $b^{(A)}, b^{(B)}$ are jointly uniform and independent of the sequence. Conditional on (A, B, R), both lifts using τ are uniform modulo P. Translating by the known products and then adding $(P/2)\mu_\beta$ preserves this distribution. Finally, floor rounding maps each component to a uniform quotient. This distribution remains uniform after conditioning on the published key, messages, and adversary state. □

The simulator needs no rank selection, search for zero errors, dual filter, or likelihood normalization. Its availability does not, however, justify the hardness assumption on the HB data it receives.

7 Conditional IND-CPA security

7.1 Game and conventions

The game generates an honest key and sends pk to a first stage $\mathcal{A}_1$. It returns $(\mu_0, \mu_1, \mathrm{st})$ with two messages in $\{0,1\}^\ell$. The challenger draws a fair β, encrypts μ_β, and sends the ciphertext and state to $\mathcal{A}_2$, which returns $\widehat{\beta}$. Define

$$\mathrm{Adv}_{\mathrm{CPA}}(\mathcal{A}) = \left|\Pr[\widehat{\beta} = \beta] - \frac{1}{2}\right|. \tag{7.1}$$

Encryption is public: honest encryption-oracle queries can be simulated locally with fresh randomness. No decryption oracle, memory leakage, or malicious key generation is included. The model is classical and standard. It uses neither a random oracle nor a short seed to expand the public sequence.

Theorem 7.1 (CPA reduction)**.** *Fix a parameter family satisfying* (3.3)–(3.4) *and an efficiently samplable distribution* χ*. For every probabilistic polynomial-time adversary* $\mathcal{A}$ *against* NOISE-CPA-LIFT*, there exist two distinguishers* $\mathcal{B}_0, \mathcal{B}_1$ *for the same problem* $\mathsf{HB}(M, d, 2n, \chi)$*, with polynomial overhead, such that*

$$\mathrm{Adv}_{\mathrm{CPA}}(\mathcal{A}) \le \mathrm{Adv}_{\mathsf{HB}}(\mathcal{B}_0) + \mathrm{Adv}_{\mathsf{HB}}(\mathcal{B}_1). \tag{7.2}$$

If the assumption in Definition 3.2 holds and correctness is perfect or has negligible failure probability, then NOISE-CPA-LIFT *is IND-CPA for this family.*

7.2 Proof through two transitions

Proof. Consider the following games, and let p_j denote the adversary's success probability in each game.

Game G_0: real execution. The key and ciphertext are produced by algorithms 1 and 2. Thus $\mathrm{Adv}_{\mathrm{CPA}}(\mathcal{A}) = |p_0 - 1/2|$.

Game G_1: ideal public observation. Keep the upper-half public sequence A, but draw U uniformly on $\mathbb{Z}_H^n$, independently of A. The adversary chooses its messages after receiving this key. Encryption remains the real algorithm, with fresh Y, ρ, E_1', E_2'. The simulator in section 6.4, which returns the indicator of the adversary's success, reproduces G_0 on a real challenge and G_1 on a null challenge. By definition,

$$|p_0 - p_1| = \mathrm{Adv}_{\mathsf{HB}}(\mathcal{B}_0).$$

Decryption under the ideal key is unnecessary.

Game G_2: jointly ideal ciphertext. Keep the key from G_1, but replace the challenge by an independent uniform element of $\mathbb{Z}_H^n \times \mathbb{Z}_H^\ell$. Proposition 6.5 establishes that the simulator in section 6.5 reproduces G_1 on a real HB challenge and G_2 on a null HB challenge. Hence

$$|p_1 - p_2| = \mathrm{Adv}_{\mathsf{HB}}(\mathcal{B}_1).$$

In G_2, the challenge does not depend on β, even conditional on the chosen messages and the adversary's state. Thus $p_2 = 1/2$. The triangle inequality proves (7.2).

The additional computations are sampling operations and Hankel products of polynomial dimensions. Runtime includes the cost of the sampler for χ. Neither simulator receives the HB challenge secret. Correctness is addressed independently in section 5; it is not used to hide the challenge bit in the games. □

Table 4: Summary of the games and the direction of reduction.

Game	Public key	Challenge	Justification
G_0	Honest (A, U)	$\mathsf{Enc}(pk, \mu_\beta)$	Real game of the specification.
G_1	Upper-half A, uniform U	Unchanged $\mathsf{Enc}(pk, \mu_\beta)$	First HB challenge, prefix, and exact mask.
G_2	Same key distribution as G_1	Jointly uniform (V, W)	Second HB challenge, two segments, and one common secret.

7.3 Resources, samplers, and multiple challenges

The naive reference algorithms require $O(nd)$ integer multiplications for key generation, $O((n+\ell)d)$ for encryption, and $O(\ell d)$ for decryption. The reductions stay within these orders, in addition to the adversary's runtime and sampling costs. Bit complexity depends on p and the sizes of the errors; modular operations are not treated as free.

If $\epsilon_{\mathsf{HB}}(T)$ bounds the advantage of all HB distinguishers with resources T, (7.2) gives the conservative bound $2\epsilon_{\mathsf{HB}}(T')$, where T' includes the simulation. This notation does not justify substituting a heuristic attack cost on a different distribution for ϵ_{HB}.

With approximate samplers, distances must be measured for each complete experiment. If the two simulated endpoints of each transition contribute at most ε_0 and ε_1, respectively, to their total distances, the bound becomes

$$\mathrm{Adv}_{\mathrm{CPA}} \le \mathrm{Adv}_{\mathsf{HB}}(\mathcal{B}_0) + \mathrm{Adv}_{\mathsf{HB}}(\mathcal{B}_1) + \varepsilon_0 + \varepsilon_1.$$

The sum accounts for the simulations actually used, not merely a scalar error without its sampling multiplicity. Multiple left–right challenges require an additional hybrid argument and incur the corresponding number of transitions; the present theorem is stated for one challenge.

7.4 An encrypted key: a corollary, not a premise

Corollary 7.2 (Joint secrecy of an encrypted key)**.** *Draw $K \leftarrow\$ \mathsf{U}(\{0,1\}^\ell)$ and publish $C = \mathsf{Enc}(pk, K)$. A distinguisher between (pk, C, K) and (pk, C, R), where R is independent and uniform and C always encrypts a uniform key, has acceptance gap at most*

$$2\big(\mathrm{Adv}_{\mathsf{HB}}(\mathcal{B}_0) + \mathrm{Adv}_{\mathsf{HB}}(\mathcal{B}_1)\big)$$

for appropriate reductions.

Proof. Choose two independent keys K_0, K_1 as the CPA messages. After receiving the challenge $C = \mathsf{Enc}(pk, K_\beta)$, give (pk, C, K_0) to the distinguisher. For $\beta = 0$, it sees the actual key; for $\beta = 1$, the supplied key is independent of the ciphertext. An acceptance gap a gives CPA bias $a/2$, up to sign. Apply theorem 7.1. □

This minimal mechanism is not a Fujisaki–Okamoto transformation and is not a CCA-secure KEM. Its correctness is that of the encryption scheme. No assumption of an "already secret key" was used to establish the main theorem.

8 Sizes, costs, and reproducible verification

8.1 Compact encoding without an additional assumption

The parameters (p, q, d, ℓ, χ) are assumed known to both parties and are not included in the sizes below. A profile identifier, protocol wrapper, public key optionally copied into a secret-key

structure, and file metadata would incur additional costs.

Each coefficient $A_i \in [M, P)$ has its high bit fixed to one. Storing $A_i - M$ requires $p - 1$ bits. Each coefficient of $U, V, W \in \mathbb{Z}_H$ requires $p - q$ bits. Finally, $X_j - Q/2$ is stored in $q - 1$ bits. Fields are concatenated without intermediate alignment, with zero padding, if needed, only at the end of each object.

Proposition 8.1 (Exact sizes of a reversible encoding)**.** *The public-key, secret-key, and ciphertext payloads contain, respectively,*

$$\begin{aligned} B_{pk} &= (2d + \ell - 2)(p - 1) + (d + \ell - 1)(p - q), \\ B_{sk} &= d(q - 1), \\ B_{ct} &= (d + 2\ell - 1)(p - q). \end{aligned} \tag{8.1}$$

Their byte lengths are $\lceil B_{pk}/8 \rceil$*,* $\lceil B_{sk}/8 \rceil$*, and* $\lceil B_{ct}/8 \rceil$*.*

Proof. There are m coefficients in A, n in U, d in X, and $n + \ell$ in the ciphertext. The maps $A_i \mapsto A_i - M$ and $X_j \mapsto X_j - Q/2$ are bijections onto the stated binary alphabets. The product encoding therefore has the sizes in (8.1) and an exact inverse. Padding bits must be zero; no information is assigned to them. □

This representation does not replace the sequence by a seed. It invokes no pseudorandom-generator indistinguishability assumption. The distributions and CPA theorem are preserved by this injective encoding: a distinguisher for binary representations can be composed with the encoder, and conversely with the canonical decoder on valid objects.

8.2 Illustrative profiles and correctness margins

Table 5: Compact sizes in bytes and CBD1 correctness. No profile is certified secure.

d	ℓ	p	q	pk	sk	ct	T_1	$P/4$
4	2	12	3	17	1	8	570	1024
16	16	14	3	118	4	65	2250	4096
16	8	16	4	106	6	47	8658	16384
128	128	17	3	1211	32	671	17930	32768
256	256	18	3	2586	64	1439	35850	65536

All five profiles satisfy the deterministic correctness bound with $b = 1$. The first three are those used in the reference's functional tests. The last two illustrate only the size and margin formulas; they are not proposed "128-bit" or "256-bit" security profiles. For example, $d = \ell = 128$ gives $m = 382$, $n = 255$, and

$$B_{pk} = 382 \cdot 16 + 255 \cdot 14 = 9682 \text{ bits},$$

hence 1211 bytes after final padding. The secret has a 256-bit encoding, but its length is not a confidentiality certificate.

A simple functional family takes $d = \ell = \lambda \geq 2$, $Q = 8$, $\chi = \mathsf{CBD}_1$, and the smallest power of two P strictly greater than $4T_1$. Its parameters and correctness requirements are polynomially bounded. This observation establishes only the existence of correct polynomial-size profiles. It does not establish the HB assumption for that family.

8.3 Reference implementation and verification methods

The accompanying public reproducibility archive contains a Python reference implementation with no external dependencies for the CBD1 profile. Generation functions use `secrets.SystemRandom` by default; tests use deterministic generators explicitly reserved for reproducibility. The implementation is not constant-time. It must not be used to protect real data.

The base verifier reconstructs the Hankel product through a polynomial convolution with a reversed secret, independently of the reference multiplication routine. Finite distributions are computed with integer counters and rational fractions, not estimated by Monte Carlo frequencies. Larger randomized tests are identified separately and are not described as exhaustive.

The public verifiers were rerun for A5 v1.1. The primary suite checks the reference arithmetic and exact finite distributions; the independent suite reconstructs selected laws without importing the reference arithmetic module. A separate deterministic encoding regression, `verify_a5_encoding.py`, supplies the exact codec coverage reported below and writes `A5_ENCODING_RESULTS.json`. The canonical encoding tests do not change the encryption algorithms or their samplers.

Table 6: Executed checks; categories are not summed as independent instances.

Type of check	Scope and result
Binary-to-positive mask	Six finite bijections; exact multiplicity one.
Scalar and uniform lifts	Five pairs (M, Q), with every offset in the tested domains.
Complete scalar games	Two messages; exact real-world equality and a uniform null world over 64 transcripts.
Conditional rectangular games	24 fixed publics, four messages each; 96 joint laws over 32 outputs, compared exactly.
Double product	4096 exhaustive integer identities.
Simulators in dimensions 2, 3, 4, 8	1000 identity couplings; not an enumeration of every distribution.
Functional correctness	3000 cases and 300 encryptions with a reused key; margins satisfied.
Decoding and malleability	4454 points within the margin; 100 ciphertext modifications tested.
Canonical encoding regression	1500 round trips; 2500 noncanonical-format rejections; 60 encoded encryption/decryption cycles.

The encoding row is reproduced by `verify_a5_encoding.py`; the deterministic output is `A5_ENCODING_RESULTS.json`. The full package wrapper executes this regression and compares the regenerated JSON with the delivered result.

8.4 Negative controls

The scalar profile $(d, \ell, P, Q) = (1, 1, 16, 8)$ lies outside the dimensions of the asymptotic theorem and does not satisfy our correctness condition. It serves only to enumerate the simulator distributions completely. For either message, removing randomized lifting from the ideal game produces an exact distance of 2295/32768 from the correct distribution. Removing the final noise produces a distance of 153/8192.

These witnesses do not attack every construction lacking those elements. They establish the narrower claim that these steps cannot be removed while citing our simulation proof unchanged. Small-modulus rectangular profiles used for exact game comparisons carry the same correctness caveat.

9 Cryptographic limitations and open obligations

9.1 Source hardness remains external

The CPA theorem can be read as an adversary transformation: an adversary against encryption yields two distinguishers for the defined HB problem. This implication is useful even before the assumption is founded, but it does not justify declaring the scheme secure in practice. An assumption can be precise yet false for particular parameters.

Further work must study exactly the $N + d - 1$-coefficient sequence, window count $N = 2(d + \ell - 1)$, binary secret, and noise χ modulo M. It should cover primal and dual attacks, algebraic attacks, and projections modulo divisors of M, without treating Hankel rows as independent samples. An estimator parameterized for general LWE does not become a certificate for this structure by merely flattening the representation.

9.2 Statistical distance and computational hardness

The real distribution can be statistically far from uniform while remaining difficult to distinguish efficiently. For example, if χ has at most K values modulo M, then for each fixed public sequence the support of a real HB output contains at most $2^d K^N$ elements. The computationally unbounded support-membership test gives

$$\mathrm{TV}(\mathsf{HB}_1, \mathsf{HB}_0) \geq \max\{0, 1 - 2^d (K/M)^N\}. \tag{9.1}$$

The proof counts possible secret–error pairs and then averages over the public sequence. Membership testing may require exponential enumeration, so (9.1) is not a polynomial-time attack. This distinction explains why normalization checks do not establish the pseudorandomness assumption.

9.3 Malleability and lack of CCA2 security

Proposition 9.1 (Flipping a bit through a related ciphertext)**.** *Under the strict correctness margin, replacing coordinate W_j of an honest ciphertext by $[W_j + H/2]_H$ flips exactly decrypted bit j, leaving all other bits unchanged. The unwrapped scheme is not IND-CCA2 secure.*

Proof. The modification adds $Q(H/2) = P/2$ to the corresponding decryption residue modulo P. The two centers are exchanged, and the strict margin excludes ties. The modified input is distinct from the challenge and remains in the canonical domain. A CCA2 adversary can therefore submit it to the decryption oracle and invert the known bit to recover the original message. With only negligibly failing correctness, the attack succeeds at least on the correctness event. □

This result does not contradict theorem 7.1. CPA does not allow decryption queries. A simple format check does not block the attack. No CCA transformation is included here; such an extension would require a new specification and proof.

9.4 Implementation, public generation, and parameters

The compact representation in section 8 is lossless. By contrast, replacing the independent coefficient sequence by the expansion of a short seed would change the generation model and require a separate analysis. The present theorem includes no ideal oracle for such an expansion. Errors, secrets, and fresh remainders must also follow their stated distributions; substituting a more convenient sampler is not silently covered.

The reference addresses neither side channels, faults, secure secret erasure, nor concurrent execution. Arbitrary-precision integers and Python branches do not provide the guarantees

required by a robust implementation. Theoretical storage sizes are not the RAM consumption of a Python program. The archive contains no cryptanalysis at security-scale dimensions or dedicated-hardware latency measurements.

9.5 Priorities for independent review

A reviewer should first check the second simulator's window selection, the conditional distribution of its internal lift after U is published, the indices in the double-product identity, and the advantage conventions. The reviewer should then compare the HB assumption with distributions in earlier publications and assess whether a parameter family can be both correct and hard. The accompanying programs facilitate these checks but are neither independent peer review nor proof-assistant verification.

10 Conclusion

We defined NOISE-CPA-LIFT as a complete encryption scheme around a precisely distributed MODDIV-NOISE public observation. Correctness follows from a double-Hankel-product identity and a deterministic margin retaining the actual remainders. Randomized quotient lifting then allows a joint ciphertext simulation from two segments of the same HB-LWE challenge with a common ephemeral secret. These ingredients yield a classical standard-model IND-CPA reduction with two exact transitions.

The article does not complete the hardness connection from standard LWE to the required binary-secret Hankel source. It recommends no concrete security profile. Its positive conclusion is conditional and circumscribed: if the specified HB family is hard and the required correctness holds, then the described construction is IND-CPA. The malleability of the unwrapped scheme explicitly limits this claim to CPA.

Priorities are independent cryptanalysis of the source and construction, parameter studies supported by a founded assumption, and comparison of the contribution with the literature. The executable reference and its encodings support that evaluation without replacing the remaining proof obligations.

Assistance disclosure and availability

Conversational AI tools, including ChatGPT, were used to explore approaches, prepare proofs, write and check programs, and draft and translate this manuscript. This assistance is neither independent scientific review, formal verification, nor institutional validation. The text is prepared for author approval before any submission. No external review is claimed in this version.

The accompanying reproducibility archive contains the LaTeX source, a public source list, Python reference implementation, two verifiers, a synthetic known-answer test, deterministic results, and integrity hashes. Third-party publications are cited but are not redistributed. No internal chats, audits, mission reports, or private research notes are part of the public scientific basis of this article. The programs are not intended for deployment.

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A Complete pseudocode for the two reductions

This appendix makes the simulator interfaces explicit. The CPA adversary is split into $\mathcal{A}_1$ and $\mathcal{A}_2$. Local encryptions performed by or for the adversary use only the public key and fresh randomness. The two reductions are distinct; each receives one challenge from the same HB family.

Algorithm 4. $\mathcal{B}_0^{\mathcal{A}}(a, b)$: transition $G_0 \leftrightarrow G_1$

Input: Challenge $a \in \mathbb{Z}_M^{N+d-1}$, $b \in \mathbb{Z}_M^N$; public parameters.
$A \leftarrow a_{0:m} + M\mathbf{1}$;
$R \leftarrow\$\ \mathsf{U}(\mathcal{R}_Q^d)$; $\tau_0 \leftarrow\$\ \mathsf{CBD}_1^{\otimes n}$;
$U \leftarrow F_Q(2b_{0:n} + \mathsf{H}_n(A)R + \tau_0)$;
$pk \leftarrow (A, U)$;
$(\mu_0, \mu_1, \mathrm{st}) \leftarrow \mathcal{A}_1(pk)$;
$\beta \leftarrow\$\ \mathsf{U}(\{0,1\})$; $ct \leftarrow \mathsf{Enc}(pk, \mu_\beta)$;
$\widehat{\beta} \leftarrow \mathcal{A}_2(\mathrm{st}, ct)$;
return $\mathbf{1}\{\widehat{\beta} = \beta\}$.

Algorithm 5. $\mathcal{B}_1^{\mathcal{A}}(a, b)$: transition $G_1 \leftrightarrow G_2$

Input: Challenge from the same family $\mathsf{HB}(M, d, N, \chi)$.
$A \leftarrow a_{0:m} + M\mathbf{1}$; $\epsilon \leftarrow\$\ \mathsf{U}(\{0,1\}^n)$;
$B \leftarrow a_{m:m+n} + M\epsilon$; $U \leftarrow F_Q(B)$;
$pk \leftarrow (A, U)$; $(\mu_0, \mu_1, \mathrm{st}) \leftarrow \mathcal{A}_1(pk)$;
$\beta \leftarrow\$\ \mathsf{U}(\{0,1\})$; $R \leftarrow\$\ \mathsf{U}(\mathcal{R}_Q^d)$;
$\tau_1 \leftarrow\$\ \mathsf{CBD}_1^{\otimes n}$; $\tau_2 \leftarrow\$\ \mathsf{CBD}_1^{\otimes \ell}$;
$V \leftarrow F_Q(2b_{0:n} + \mathsf{H}_n(A)R + \tau_1)$;
$W \leftarrow F_Q(2b_{m:m+\ell} + \mathsf{H}_\ell(B)R + \tau_2 + (P/2)\mu_\beta)$;
$\widehat{\beta} \leftarrow \mathcal{A}_2(\mathrm{st}, (V, W))$;
return $\mathbf{1}\{\widehat{\beta} = \beta\}$.

A.1 A nonscalar indexing example

Take $d = 3$, $\ell = 2$. Then $n = 4$, $m = 6$, $N = 8$, and the source sequence has ten coefficients $a_0, \ldots, a_9$. The first segment is $(a_0, \ldots, a_5)$; its four observations start at indices $0, 1, 2, 3$. Observations b_4, b_5 cross the boundary and are discarded. The second segment is $(a_6, \ldots, a_9)$; its two observations start at indices $6, 7$. The errors in $b_0, b_1, b_2, b_3, b_6, b_7$ are distinct, whereas all observations depend on the same three-bit secret.

Discarded observations could help an adversary against the HB assumption; their presence does not make that assumption less demanding. The reduction does not transmit them, so they may be marginalized out. The reuse of public coefficients within each segment is preserved exactly by the two product computations.

A.2 Why presampling the lift reveals nothing

In the second simulator, B is chosen before the adversary returns its messages. The adversary sees only $U = F_Q(B)$. For every $U = u$, the preimage set contains Q^n equally likely values. The adversary's state is determined by pk and its own coins, so it provides no additional access to the remainders of B beyond u.

This is a conditional-distribution property, not a claim that B is independent of U. On the contrary, their relation $F_Q(B) = U$ is deterministic and necessary. Forgetting that relation would give an incorrect simulator even if B and U separately had the correct marginals.

B Decoding, noise, and encoding details

B.1 Exact table of the lifted CBD1 error

In the reference, e and τ are independent differences of fair bits. The convolution defining $E' = 2e + \tau$ is:

Value of E'	-3	-2	-1	0	1	2	3
Weight out of 16	1	2	3	4	3	2	1

Explicit errors at different positions follow this same distribution with independent draws. When residues coincide under a small modulus, their weights are added in the counters. Correctness uses the integer lifts above.

B.2 Decoder boundaries

The decoder chooses one if and only if the center $P/2$ is strictly closer. For a canonical residue z, this is equivalent to $P/4 < z < 3P/4$. Residues $P/4$ and $3P/4$ are decoded as zero by convention. Hence, for an integer error $|\delta| < P/4$, the representatives of $[(P/2)\mu + \delta]_P$ always decode to μ, including when $\delta < 0$.

This definition specifies behavior for every valid ciphertext, not only those satisfying the margin. It provides no confidentiality guarantee concerning the errors or validity of a maliciously submitted ciphertext.

B.3 Canonical binary format

Coefficients are ordered by increasing index. Within each field, the least significant bit is placed first. The concatenation is interpreted as an integer whose byte encoding is little-endian. Unused bits in the final byte are zero.

For the public key, concatenate the m values $A_i - M$ in $p - 1$ bits, followed by the n values U_i in $p - q$ bits. For the secret key, concatenate $X_j - Q/2$ in $q - 1$ bits. For the ciphertext, concatenate V then W, using $p - q$ bits per coefficient. The decoder checks exact length and zero padding; it rejects extra length even if the appended bytes are zero.

The parameters are not encoded in these payloads. A protocol using several profiles must identify their parameters and samplers unambiguously through an external mechanism. No such negotiation protocol is specified here.

C Reproducibility package guide

C.1 Organization and commands

The package contains the PDF, LaTeX source, public source list, programs, deterministic results, and integrity hashes. The scripts use only Python's standard library. Results can be reproduced from the stated seeds; test counts and distribution comparisons do not depend on execution timings or on environment metadata.

From the extracted package directory, run:

```
python3 verify_a5.py --output A5_RESULTS_RECHECK.json
python3 verify_a5_independent.py --output A5_INDEPENDENT_RECHECK.json
python3 a5_kat.py
```

`verify_a5.py` regenerates the primary exact-check result file.

`verify_a5_independent.py` is a separately written checker that imports neither `a5_noise.py` nor `a5_codec.py`; it reconstructs selected joint laws, the corrected deferred-lift condition, and finite counterexamples independently.

`a5_kat.py` produces `A5_KAT.json`, a synthetic test vector with deterministic coins. This vector intentionally exposes a test key; it has no security value.

C.2 Reference interface

```
from a5_noise import Params, keygen, encrypt, decrypt
from a5_codec import pack_pk, unpack_pk, pack_ct, unpack_ct

p = Params(P=4096, Q=8, d=4, ell=2)
assert p.correctness_certified(raw_error_bound=1)
pk, sk = keygen(p)
message = (1, 0)

pk_wire = pack_pk(pk)
pk_decoded = unpack_pk(p, pk_wire)
ct = encrypt(pk_decoded, message)
ct_wire = pack_ct(p, ct)
ct_decoded = unpack_ct(p, ct_wire)
assert decrypt(sk, ct_decoded) == message

assert len(pk_wire) == 17
assert len(ct_wire) == 8
# Functional example only; not recommended for deployment.
```

C.3 Status of each verification family

Exact game comparisons sum over secrets and errors only in the small verifiers; enumeration is not part of the reductions. Functions `hop1_public`, `hop2_public`, and `hop2_challenge` do not accept the source secret as an input. Synthetic challenge generators know a secret because they create test instances, not because a simulator has access to it.

Functional tests do not measure a cryptographic-scale failure probability. Their purpose is to compare the implementation with the deterministic identity of lemma 5.2. Theorem 5.3 provides the mathematical guarantee for bounded-noise profiles. Counterexamples without lifting or final noise check the simulation obligation, rather than proving every such variant insecure.

The file `SHA256SUMS.txt` identifies the public artifacts in this edition. It checks their relative integrity, not an author's identity or an official publication. No font files or copies of third-party articles are distributed.

C.4 Public-version traceability

This public version is self-contained: the construction, source experiment, correctness proof, two exact reductions, encoding rules, CCA2 attack, and limitations are all stated in the article. The reproducibility archive contains only public-facing source and verification artifacts. Exploratory research paths not formalized in this article are not premises of the CPA theorem. Producing this document announces no submission acceptance, external validation, or exhaustive priority determination.